

# Xiao FU (付潇)

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## EDUCATION

### The Chinese University of Hong Kong (CUHK)

Ph.D. in Information Engineering, Multimedia Laboratory (MMLab)

- **Advisor:** Prof. *Dahua Lin*

Hong Kong SAR

2023 – Present

### Zhejiang University (ZJU)

B.Eng. (Honor) in Information Engineering

- **Advisor:** Prof. *Yiyi Liao*, **GPA:** 3.96/4.00 (Top 1%)

Hangzhou, China

2018 – 2022

## SELECTED PUBLICATION

- **Cosmos 3: Omnimodal World Models for Physical AI**

Xiao Fu (Core Contributor): Long-Context Interactivity, Data Curation, Synthetic Data Generation

NVIDIA Cosmos WhitePaper, 2026 | [website](#) | [product](#) | [code \(11K+ stars\)](#)

- **Plenoptic Video Generation**

Xiao Fu, Shitao Tang, Min Shi, Xian Liu, Jinwei Gu, Ming-Yu Liu, Dahua Lin, Chen-Hsuan Lin

Computer Vision and Pattern Recognition (CVPR), 2026 | [website](#) | [code](#)

- **Cosmos 2.5: World Simulation with Video Foundation Models for Physical AI**

Xiao Fu (Core Contributor): Multi-Camera & Interactive Video Generation, Multi-View Robotic Transfer

NVIDIA Cosmos WhitePaper, 2025 | [website](#) | [code \(predict2.5\)](#) | [code \(transfer2.5\)](#)

- **Learning Video Generation for Robotic Manipulation with Collaborative Trajectory Control**

Xiao Fu, Xintao Wang, Xian Liu, Jianhong Bai, Runsen Xu, Pengfei Wan, Di Zhang, Dahua Lin

International Conference on Learning Representations (ICLR), 2026 | [website](#) | [code \(100+ stars\)](#)

- **3DTrajMaster: Mastering 3D Trajectory for Multi-Entity Motion in Video Generation**

Xiao Fu, Xian Liu, Xintao Wang, Sida Peng, Menghan Xia, Xiaoyu Shi, Ziyang Yuan, Pengfei Wan, Di Zhang, Dahua Lin

International Conference on Learning Representations (ICLR), 2025 | [website](#) | [dataset](#) | [code \(300+ stars\)](#)

- **PanopticNeRF-360: Panoramic 3D-to-2D Label Transfer in Urban Scenes**

Xiao Fu, Shangzhan Zhang, Tianrun Chen, Yichong Lu, Xiaowei Zhou, Andreas Geiger, Yiyi Liao

IEEE Transactions on Pattern Analysis and Machine Intelligence (T-PAMI), 2025 | [website](#) | [code \(200+ stars\)](#)

- **GeoWizard: Unleashing the Diffusion Priors for 3D Geometry Estimation from a Single Image**

Xiao Fu, Wei Yin, Mu Hu, Kaixuan Wang, Yuxin Ma, Ping Tan, Shaojie Shen, Dahua Lin, Xiaoxiao Long

European Conference on Computer Vision (ECCV), 2024 | [website](#) | [code \(900+ stars\)](#) (**Citation: 300+**)

- **Panoptic NeRF: 3D-to-2D Label Transfer for Panoptic Urban Scene Segmentation**

Xiao Fu, Shangzhan Zhang, Tianrun Chen, Yichong Lu, Lanyun Zhu, Xiaowei Zhou, Andreas Geiger, Yiyi Liao

International Conference on 3D Vision (3DV), 2022 | [website](#) | [code \(200+ stars\)](#) (**Citation: 200+**)

- **ReCamMaster: Camera-Controlled Generative Rendering from A Single Video**

Jianhong Bai, Menghan Xia, Xiao Fu, Xintao Wang, Lianrui Mu, Jinwen Cao, Zuozhu Liu, Haoji Hu, Xiang Bai, Pengfei Wan, Di Zhang

International Conference on Computer Vision (ICCV), 2025 (**Best Paper Award Candidate, Top 13/11,239**)

- **OmniObject3D: Large-Vocabulary 3D Object Dataset for Realistic Perception, Reconstruction and Generation**

Tong Wu, Jiarui Zhang, Xiao Fu, Yuxin Wang, Jiawei Ren, Liang Pan, Wayne Wu, Lei Yang, Jiaqi Wang, Chen Qian, Dahua Lin, Ziwei Liu

Computer Vision and Pattern Recognition (CVPR), 2023 (**Best Paper Award Candidate, Top 12/9,155**)

## WORKING EXPERIENCE

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### Cosmos Lab, NVIDIA Research

Research Scientist Intern. Advisor : Dr. *Ming-Yu Liu* (VP of Research)

Topic: Cosmos World Foundation Models for Physical AI

Santa Clara, United States

May 2025 – May 2026

### Kling Team, Kuaishou VGI Lab

Research Intern. Advisor : Dr. *Xintao Wang*

Topic: Controllable Video Generation

Shenzhen, China

May 2024 – May 2025

### Shanghai AI Innovation Center

Associate Researcher. Advisor : Prof. *Dahua Lin*

Topic: OmniObject3D: A Large-Vocabulary and Realistic 3D Object Dataset

Shanghai, China

Jun. 2022 – Jun. 2023

### Max Planck Institute for Intelligent Systems

Researcher Assistant. Advisor : Prof. *Andreas Geiger* & Prof. *Yiyi Liao*

Topic: Panoptic Neural Radiance Field for 3D-2D Label Transfer

Tübingen, Germany (remote)

Sept. 2021 – Mar. 2022

## INVITED TALK & WORKSHOPS

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- **NVIDIA Research:** Multi-Camera Video Generation for Embodied Intelligence 2025
- **AnySyn3D Webinar:** 3D Interactive Video Generation 2024
- **AI for 3D Content Creation Workshop at ICCV 2023:** OmniObject3D Challenge 2023
- **Shenlan Open Courses:** Panoramic Label Rendering for Autonomous Driving Simulator 2023

## HONORS & AWARDS

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- ICCV Best Paper Award Candidate 2025
- CVPR Best Paper Award Candidate 2023
- Hong Kong PhD Fellowship Scheme (HKPFS) (**Top 300 Ph.D. students in Hong Kong**) 2023
- CUHK Vice-Chancellor HKPFS Scholarship 2023
- Outstanding Graduation Thesis Award of Zhejiang University 2022
- National Scholarship, Ministry of Education of P.R. China (**Top 0.2% across China**) 2020, 2021
- ZJU ISEE Excellent Student Award (**Dean's Honor Undergraduate**) 2021
- ZJU ISEE Yuelun Alumni Scholarship (**Dean's Award for Academic Contest**) 2020
- Finalist Winner, Mathematical Contest in Modeling (MCM/ICM) (**Top 2% out of 20,954 teams**) 2020
- Gold Prize, 7<sup>th</sup> "Internet Plus" College Student Innovation and Entrepreneurship Contest (1<sup>st</sup> author) 2021